

The University of California, Santa Cruz

PHYS 133: Intermediate Laboratory

Lab 2: Absorption of Gamma Rays by Matter

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Absorption of Gamma Rays by Matter

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Abstract

In this experiment, we measured and derived many important quantities relating to the detection of gamma rays and their attenuation by lead, including the amplitude threshold at which our counter was able to register gamma rays; the intrinsic net resistance, intrinsic total capacitance, and charge of the counter circuit; the similarity of the counting statistics to a Poisson distribution; the time it takes for the counter to register counts after an influx of counts (dead time); and the absorption mass coefficient of lead. We first measured the amplitudes of the pulses from the counter as well as the count rates at different counter voltages to find a counter threshold amplitude between 744 ± 1 mV and 920 ± 1 mV. Then, we measured the amplitude and time constant with and without a known capacitor attached in parallel to calculate the intrinsic net resistance, intrinsic total capacitance, and charge on the counter's central wire, which we concluded to be 509 ± 51 MΩ, 1020 ± 103 pF, and 1.06 ± 0.08 nC respectively. Our histogram of counts during 100 one second intervals ended up varying a not insignificant amount from the Poisson distribution, leading us to believe that either our data was not perfectly modeled by a Poisson distribution, or that we managed to gather a statistically improbable set of data by chance. Next, we directly measured the dead time using our oscilloscope, giving us an estimate of 290 ± 4 μs. We then indirectly calculated the dead time and got a result of 200 ± 75 μs. Finally, we calculated the absorption mass

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coefficient of lead at the energy of our gamma rays, which ended up with a result of 0.102 cm^2/g.

Keywords: Geiger-Mueller Counter, Counter Threshold Amplitude, Dead Time, Absorption Mass Coefficient

1 Introduction

The purpose of this experiment is to measure how the rate of gamma ray detection is affected by attenuation by sheets of lead and determine if that rate is exponentially dependent on the thickness of the lead shielding, as predicted by theory. To understand how the gamma ray counting rate is affected by lead shielding, one must first understand how the gamma rays are detected and counted. Gamma ray detectors work by converting gamma rays into high energy electrons or electron/positron pairs through photoelectric absorption, Compton scattering, and pair production in a rare gas such as argon [1]. The electrons ionize the gas, causing more ionization as they are attracted to a central wire, so that current pulses along this wire can be measured and counted, thus giving an idea of the relative number of gamma rays entering the tube. This number of counts is therefore affected by the voltage difference between the central wire and the outer cylindrical shell of the counter.

Important to our experiment, the charge on the central wire (Q), the intrinsic total capacitance of the counter (C), and the intrinsic net resistance in the counter (R) can be calculated by attaching a known capacitor (C_k) and measuring the amplitudes of the wire's voltage (A) and the time constant (τ_{RC}) of the pulses as they come in with and without the capacitor. With these measurements, Q, C, and R can be calculated according to these equations (see Equations 1 and 2).

$$\tau_{RC} = R(C + C_k) \tag{1}$$

$$A = Q/(C + C_k) \tag{2}$$

Since the time constant without the capacitor is defined as R multiplied by C, these equations can be rewritten like this (see Equations 3 and 4).

$$\tau_{RC,cap.} = \tau_{RC,none} + RC_k \tag{3}$$

$$Q = A_{cap.}(C + C_k) \tag{4}$$

Where $\tau_{RC,cap.}$ is the time constant with the capacitor, $\tau_{RC,none}$ is the time constant without it, and $A_{cap.}$ is the pulse amplitude with the capacitor

We also want to verify that the count rate of those pulses is consistent with a Poisson distribution, which can be done by comparing the probabilities of numbers of counts in one second to the Poisson distribution function (see Equation 5).

$$P(m) = e^{-\lambda\Delta t}\frac{(\lambda\Delta t)^m}{m!} \tag{5}$$

Where P(m) is the probability of m counts in one second and $\lambda\Delta t$ is the average number of counts in one second.

Also important to our experiment is the counter dead time (τ), or the time it takes for the counter to be able to detect particles after it is discharged by other particles passing through it. The indirect method for calculating the dead time involves taking counting rates for two sources individually (n_a and n_b) and together (n_{ab}) and then using this formula (see Equation 6).

$$\tau = \frac{1}{n_{ab}}[1 - [1 - \frac{n_{ab}(n_a + n_b - n_{ab})}{2n_a n_b}]^{\frac{1}{2}}] \tag{6}$$

Finally, in measuring the absorption of gamma rays by lead, we must use the mass of the lead sheets (M), their area (a), and the count rates (I) to calculate the absorption mass coefficient of lead ($\frac{\mu}{\rho}$) with approximately 0.662 MeV gamma rays using this equation (see Equation 7) and compare it to the absorption mass coefficient in Table 5 at this energy.

$$I = I_0 e^{-\frac{\mu M}{\rho a}} \tag{7}$$

2 Apparatus and Procedure

In setting up this experiment, the main pieces of equipment involved are the Geiger-Mueller tube, which works as described in the introduction, the SPECTECH counter box, which provides the high voltage to the tube and reads the output pulses from the tube, giving us a number of counts and a count rate, and a Tektronix digital oscilloscope, which the counter box connects to to display the shapes of the pulses and their amplitudes.

Figure 1 is a schematic diagram of the setup used in this experiment

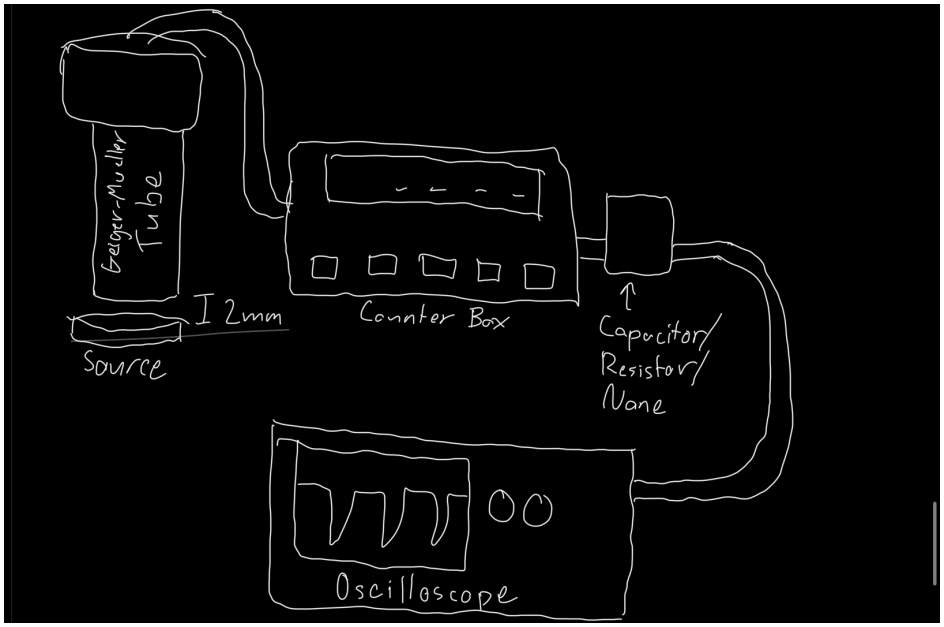


Figure 1: Schematic diagram drawn based on the setup we used for this experiment

We began by setting up the Geiger-Mueller tube for our experiment, fixing the end of the tube two millimeters directly above the gamma ray emitting side of the Caesium-137 source provided by our instructor. With this set up, we turned on the counter box and oscilloscope, positioning the output voltage line in the center of the oscilloscope screen. We then began the first part of our experiment, setting the counter high voltage to 400 Volts and gradually moving it up to 1000 Volts, measuring the pulse amplitude and count rate with each increase. This allowed us to identify the counter threshold amplitude and high voltage, which is where

the pulse amplitude becomes large enough to register a count on the counter.

Next, we attached a capacitor of capacitance C_k=2200 ± 220 pF in parallel across the input of the oscilloscope and measured the pulse amplitudes and the time constants of the pulses with and without the capacitor. From these numbers we calculated the charge on the wire (Q), the intrinsic total capacitance (C), and the intrinsic total resistance (R) using equations 3 and 4 as described in the introduction.

Then we replaced the capacitor with a resistor of resistance 10 kΩ and observed how the pulses were greatly shortened, which is helpful for later parts of this experiment, so we kept it on. With the resistor in place, we went on to verify that the count rates are consistent with a Poisson distribution. We did this by measuring the counts in 100 one-second intervals and comparing a histogram of these counts to the Poisson distribution function (Equation 5) with $\lambda\Delta t$ equal to the average number of counts in one second.

For the next part of our experiment, we obtained the counter dead time both directly and indirectly. Before we measured though, we flipped the source over to the beta and gamma side and set the counter high voltage so that the average pulse amplitude was twice the counter threshold amplitude. We measured the dead time directly by setting the oscilloscope sweep to 500 microseconds and setting a persistent trace on the screen to overlay many pulses. We then measured the time between the last full amplitude pulse and the next pulse to reach the counter threshold amplitude, giving us the dead time. To calculate it indirectly, we measured the counting rates from two sources individually (n_a and n_b) and together (n_{ab}) five times each and averaged them. With these averages, we were able to approximate the counter dead time using Equation 6.

For the final part of our experiment, measuring the absorption of gamma rays by lead, we returned to using one source and flipped it back to the gamma only side. We also raised the Geiger-Mueller tube up to 47 millimeters from the source, so we could fit all the sheets of lead we needed between the two. Again, at twice the counter threshold amplitude, we began

by moving the source far away from the tube and taking a reading of the count rate from background radiation so we can subtract this from our other count rates. Now, with the source back under the tube, we took a reading of the count rate with no lead shielding to get a starting point. From here, we began stacking on lead plates of the same area two at a time, measuring the mass and thickness of the total lead shielding with every two plates added. At every thickness of lead, we also measured the count rate. We then plotted the natural log of the count rates versus the lead mass divided by the area of the sheets, giving us a linear graph with a slope representing the absorption mass coefficient of lead for gamma rays of this energy (assumed to be 0.662 MeV). Then, we plotted the absorption mass coefficient of lead versus gamma ray energy according to Table 5 near this energy, allowing us to compare our experimental result for the absorption mass coefficient to the theoretical values for similar energies.

3 Results

3.1 *Counter Threshold Amplitude*

Below are the pulse amplitudes and count rates as the counter high voltage was raised from 400 volts to 1000 volts in 20 volt intervals. (see Table 1).

Table 1
Amplitudes and Count Rates at Varying Counter High Voltages

$HighVoltage[V]$	$Amplitude[mV]$	$CountRate[s^{-1}]$
400	30 ± 1	0
420	32 ± 1	0
440	32 ± 1	0
460	34 ± 1	0
480	36 ± 1	0
500	36 ± 1	0
520	36 ± 1	0
540	38 ± 1	0
560	38 ± 1	0
580	38 ± 1	0
600	40 ± 1	0
620	40 ± 1	0
640	40 ± 1	0
660	60 ± 1	0
680	256 ± 1	0
700	464 ± 1	0
720	744 ± 1	0
740	920 ± 1	18
760	1020 ± 1	124
780	1180 ± 1	128
800	1360 ± 1	135
820	1480 ± 1	138
840	1560 ± 1	135
860	1820 ± 1	145
880	1940 ± 1	147
900	2060 ± 1	147
920	2240 ± 1	150
940	2480 ± 1	151
960	2600 ± 1	151
980	2840 ± 1	150
1000	3000 ± 1	152

The figure below shows plots of pulse amplitude vs. counter high voltage and count rate vs. counter high voltage (see Figure 2).

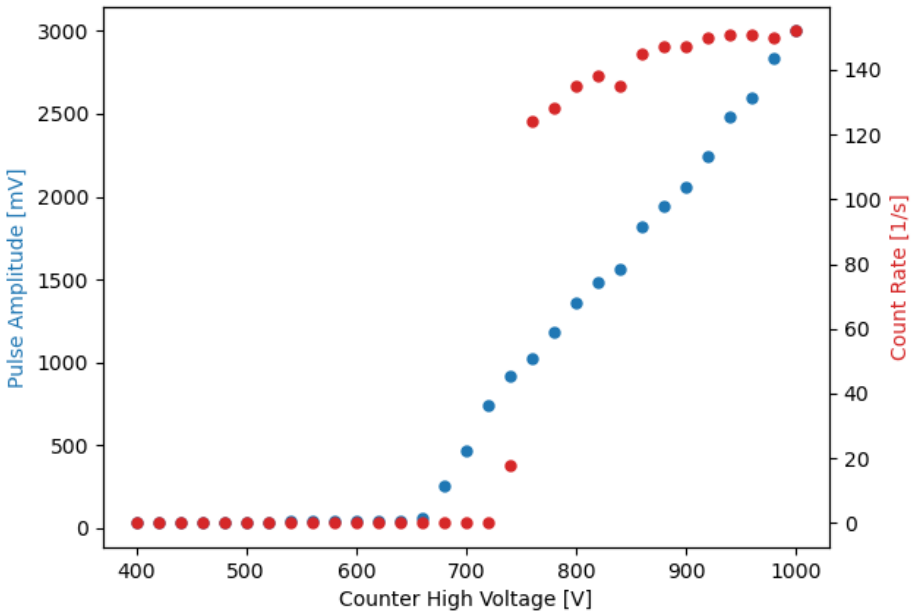


Figure 2: Pulse Amplitude (Blue) and Count Rate (Red) vs. Counter High Voltage

Table 1 and Figure 2 show that the counter threshold amplitude occurs at a counter high voltage of 740 V and the threshold amplitude is somewhere between 744 ± 1 mV and 920 ± 1 mV, which corresponds woth the claimed threshold amplitude of 800 mV.

3.2 *Calculation of Charge, Intrinsic Capacitance, and Intrinsic Resistance*

With the counter threshold amplitude measured, we moved on to calculating the charge on the central wire (Q), the intrinsic total capacitance (C), and the intrinsic net resistance (R). Below is a table containing the attached, known capacitance (C_k), the pulse amplitude (A), and the time constant (τ_{RC}) for the cases of no capacitor and the known capacitor attached (see Table 2).

Table 2
Amplitudes and Time Constants With and Without Known Capacitor

$Attached Capacitance (C_k)[pF]$	$Amplitude (A)[mV]$	$Time Constant (\tau_{RC})[s]$
0 ± 0	840 ± 1	0.52 ± 0.005
2200 ± 220	328 ± 1	1.64 ± 0.005

From this information, we were able to use equations 3 and 4 to calculate the charge (Q), intrinsic total capacitance (C), and intrinsic net resistance (R). By manipulating the equations, we calculated an intrinsic net resistance of 509 ± 51 MΩ, an intrinsic total capacitance of 1020 ± 103 pF, and a charge on the central wire of 1.06 ± 0.08 nC.

3.3 *Investigating the Counting Statistics*

Below is a histogram showing the amount of times each number of counts happened in the one hundred one second intervals. Overlayed on top of the histogram is a plot of the Poisson distribution function with $\lambda\Delta t$ equal to the average number of counts in one second, which was 10.99 (see Figure 3).

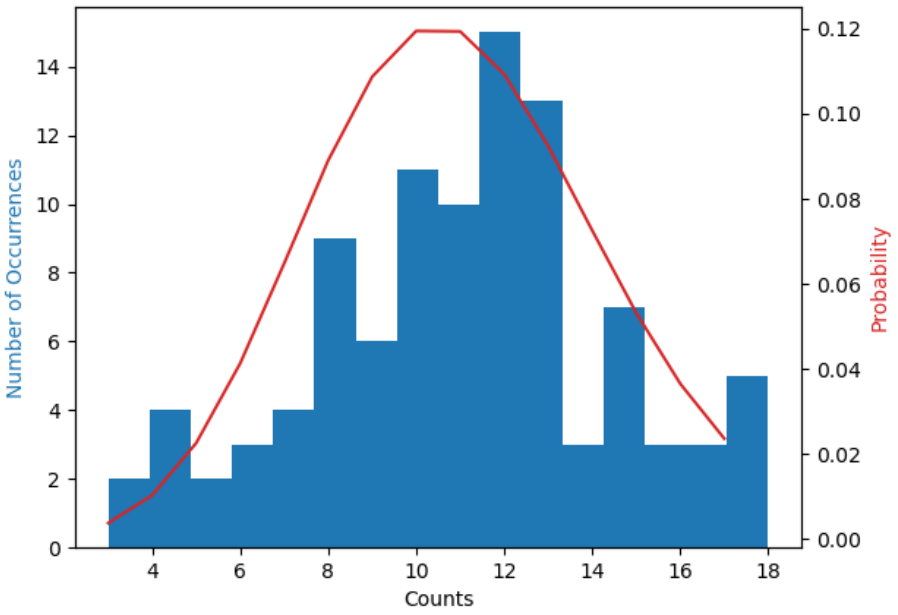


Figure 3: Histogram of Counts in One Second (Blue) with the Poisson Distribution Function (Red)

A chi-square test with eleven bins gave a result of 21.34, which is significantly larger than the number of bins, suggesting that our data was not described perfectly by a Poisson distribution function, or that we just happened to get a statistically unlikely set of data. The latter seems more likely, as the histogram seems to be following the Poisson distribution function fairly closely in some parts, but there are some strange outlier counts like fourteen and eighteen.

3.4 *Counter Dead Time*

The table below shows the counting rates from two different sources individually (n_a and n_b), as well as together (n_{ab}). At the bottom, it also shows the average of each of these quantities in each respective column (see Table 3).

Table 3
Number of Counts for Two Individual Sources and for the Combination of Both

Source A Count Rate (n_a)[s^{-1}]	Source B Count Rate (n_b)[s^{-1}]	Count Rate From Both Sources (n_{ab})[s^{-1}]
875.4 ± 26.3	897.8 ± 26.9	1209.4 ± 36.3
880.0 ± 26.4	895.0 ± 26.9	1196.5 ± 35.9
894.0 ± 26.8	883.3 ± 26.4	1207.3 ± 36.2
884.8 ± 26.5	890.0 ± 26.7	1213.7 ± 36.4
883.4 ± 26.5	885.5 ± 26.6	1209.6 ± 36.3
883.5 ± 26.5	890.3 ± 26.7	1207.3 ± 36.2

Using these averages and Equation 6, we calculated the dead time to be 0.000200 ± 0.000075 seconds, or 200 ± 75 microseconds. This is close to what we measured directly, as described in the procedure, which was 290 ± 4 microseconds, though it is important to note that the direct method is not exact and only provides an estimate of the dead time [1].

3.5 *Absorption of Gamma Rays by Lead*

Below is a table showing the mass of lead shielding being used (M), the thickness of the lead shielding (x), and the count rates at each level of shielding (I) (see Table 4). All sheets of lead used for shielding had an area of 104.04 ± 0.25 cm^2. Also, the measured background counting rate was 0.48 s^{-1}.

Table 4
Count Rates from Cs-137 Source at Different Masses and Thicknesses of Lead Shielding

Lead Mass (M) [g]	Lead Thickness (x) [mm]	Count Rate (I) [s^{-1}]
0 ± 0	0 ± 0	36.99 ± 1.1
395.1 ± 0.05	4.05 ± 0.05	29.73 ± 0.9
794.5 ± 0.05	7.71 ± 0.05	18.74 ± 0.6
1191.4 ± 0.05	11.37 ± 0.05	12.73 ± 0.4
1895.6 ± 0.05	16.91 ± 0.05	6.42 ± 0.2
2295.0 ± 0.05	20.57 ± 0.05	4.64 ± 0.1
2691.5 ± 0.05	24.23 ± 0.05	3.30 ± 0.1
3092.5 ± 0.05	27.96 ± 0.05	2.31 ± 0.07
3489.6 ± 0.05	31.86 ± 0.05	1.78 ± 0.05

Using the count rates, the lead mass (M), and the area of the sheets (a), we then plotted the natural log of the count rates versus $\frac{M}{a}$, which gave us this plot (see Figure 4).

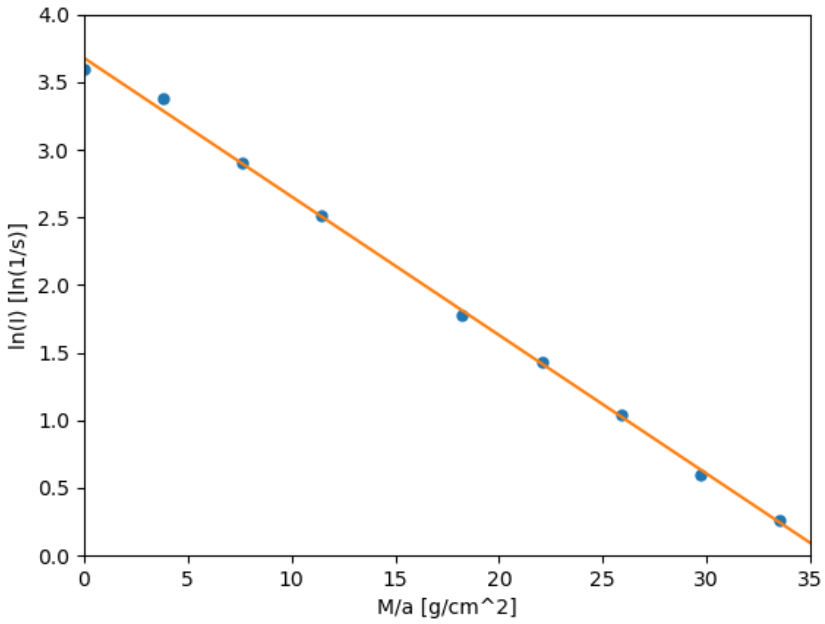


Figure 4: Natural Log of Count Rate vs. Lead Mass Divided by Sheet Area

We calculated the slope of the line of best fit to be -0.102, which means our result for

the absorption mass coefficient of lead is 0.102 cm^2/g.

The table below shows the relevant photon energies and mass coefficients of lead (see Table 5).

Table 5
Relevant Photon Energies and Mass Coefficients of Lead

$PhotonEnergy[MeV]$	$MassCoefficient(\frac{\mu}{\rho})[cm^2/g]$
0.4086	0.214
0.5108	0.149
0.6811	0.102
1.022	0.0682
1.362	0.0549

Graphing mass coefficient versus photon energy from Table 5 gave us the figure below (see Figure 5).

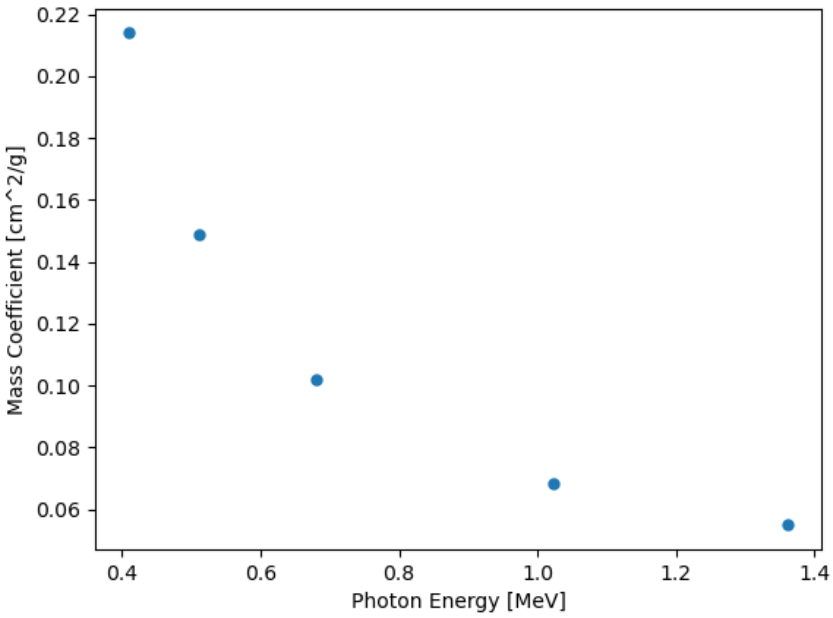


Figure 5: Absorption Mass Coefficient vs. Photon Energy

In this figure, we see that our result for the absorption mass coefficient falls well in the range of possible coefficients for gamma ray energies around 0.662 MeV.

4 Discussion and Conclusion

Overall, the results we got lined up pretty well with what was expected. We found the counter threshold amplitude to be somewhere between 744 ± 1 mV and 920 ± 1 mV, which is consistent with the counter's claimed threshold of 800 mV. The threshold also corresponded to a counter high voltage of 740 V. We found the intrinsic net resistance to be 509 ± 51 MΩ, the intrinsic total capacitance to be 1020 ± 103 pF, and the charge on the central wire to be 1.06 ± 0.08 nC. We did not have expected values for these, however, they appear to be within reason. The count rate statistics for one second intervals we measured were found to be slightly off from a typical Poisson distribution via a chi-square test, which was not expected, but the chi-square value was close enough to the number of bins used for this to have possibly been a slight statistical fluke.

Directly measuring the dead time using the oscilloscope gave us an estimate of 290 ± 4 μs, and the dead time we calculated indirectly was 200 ± 75 μs. This seems slightly off, however, it is important to note that the method for measuring the dead time directly is not exact and only provides an estimate [1], so the actual counter dead time may have been closer to the indirect calculation. Finally, in our calculation of the absorption mass coefficient of lead, we achieved a result of 0.102 cm^2/g, which would, according to Table 5, correspond to a photon energy of exactly 0.6811 MeV, though the actual number may be slightly different from that.

5 Acknowledgements

My lab partner, Arsh Sayyad, helped me gather some of the data for these experiments. Dr. Pablo Saz Parkinson and his teaching assistant, Swosti Choudhury, also provided some guidance throughout the experiments.

6 Appendix: Error Analysis

The error in the value of the intrinsic net resistance (R) was calculated using Equation 11.

$$\frac{\delta R}{\delta \tau_{RC,cap.}} = \frac{1}{C_k} \tag{8}$$

$$\frac{\delta R}{\delta \tau_{RC,none}} = -\frac{1}{C_k} \tag{9}$$

$$\frac{\delta R}{\delta C_k} = \frac{\tau_{RC,none} - \tau_{RC,cap.}}{C_k^2} \tag{10}$$

$$\sigma_R = ((\frac{\delta R}{\delta \tau_{RC,cap.}})^2 \sigma^2_{\tau_{RC,cap.}} + (\frac{\delta R}{\delta \tau_{RC,none}})^2 \sigma^2_{\tau_{RC,none}} + (\frac{\delta R}{\delta C_k})^2 \sigma^2_{C_k})^{\frac{1}{2}} \tag{11}$$

The error in the intrinsic total capacitance was calculated using Equation 14.

$$\frac{\delta C}{\delta \tau_{RC,none}} = \frac{1}{R} \tag{12}$$

$$\frac{\delta C}{\delta R} = -\frac{\tau_{RC,none}}{R^2} \tag{13}$$

$$\sigma_C = ((\frac{\delta C}{\delta \tau_{RC,none}})^2 \sigma^2_{\tau_{RC,none}} + (\frac{\delta C}{\delta R})^2 \sigma^2_R)^{\frac{1}{2}} \tag{14}$$

The error in the charge on the central wire (Q) was calculated using Equation **??**.

$$\frac{\delta Q}{\delta A_{cap.}} = C + C_k \tag{15}$$

$$\frac{\delta Q}{\delta C} = A_{cap.} \tag{16}$$

$$\frac{\delta Q}{\delta C_k} = A_{cap.} \tag{17}$$

$$\sigma_Q = ((\frac{\delta Q}{\delta A_{cap.}})^2 \sigma^2_{A_{cap.}} + (\frac{\delta Q}{\delta C})^2 \sigma^2_C + (\frac{\delta Q}{\delta C_k})^2 \sigma^2_{C_k})^{\frac{1}{2}} \tag{18}$$

The error in indirect calculation of counter dead time (τ) was calculated according to

Equation 22.

$$\frac{\delta\tau}{\delta n_a} = -(\frac{n_{ab}(n_a+n_b-n_{ab})}{4n_a^2 n_b n_{ab}} - \frac{n_{ab}}{4n_a n_b n_{ab}})(1-\frac{n_{ab}(n_a+n_b-n_{ab})}{2n_a n_b})^{-\frac{1}{2}} \tag{19}$$

$$\frac{\delta\tau}{\delta n_b} = -(\frac{n_{ab}(n_a+n_b-n_{ab})}{4n_a n_b^2 n_{ab}} - \frac{n_{ab}}{4n_a n_b n_{ab}})(1-\frac{n_{ab}(n_a+n_b-n_{ab})}{2n_a n_b})^{-\frac{1}{2}} \tag{20}$$

$$\frac{\delta\tau}{\delta n_{ab}} = (\frac{n_{ab}(n_a+n_b-n_{ab})}{4n_a n_b n_{ab}} - \frac{n_{ab}}{4n_a n_b n_{ab}})(1-\frac{n_{ab}(n_a+n_b-n_{ab})}{2n_a n_b})^{-\frac{1}{2}} - (1-(1-\frac{n_{ab}(n_a+n_b-n_{ab})}{2n_a n_b})^{\frac{1}{2}})n_{ab}^{-2} \tag{21}$$

$$\sigma_\tau = ((\frac{\delta\tau}{\delta n_a})^2\sigma_{n_a}^2 + (\frac{\delta\tau}{\delta n_b})^2\sigma_{n_b}^2 + (\frac{\delta\tau}{\delta n_{ab}})^2\sigma_{n_{ab}}^2)^{\frac{1}{2}} \tag{22}$$

References

[1] G. Brown. *Laboratory Manual: Physics 133*. 2024.